第一章　态和力学量的表象

考试内容 1

态的表象、力学量算符的矩阵表示、矩阵力学、幺正变换、Dirac 符号、线性谐振子及占有数表象

考试要求 1

1. 掌握表象变换、幺正变换、Dirac 符号，会求力学量算符的本征值和本征态，掌握线性谐振子的代数解法
2. 能综合运用矩阵力学的基本知识解决问题

1.1　态的表象

1.1.1　各表象能量本征方程

1. 坐标表象：

$$\left[-\frac{\hbar^2}{2m}\frac{\mathrm{d}^2}{\mathrm{d}x^2}+V(x)\right]\psi(x)=E\psi(x) \tag{1.1}$$

2. 动量表象：

$$\begin{aligned} -\frac{p^2}{2m}\varphi(p)+\int V_{pp'}\varphi(p')\mathrm{d}p'&=E\varphi(p)\\ V_{pp'}=\frac{1}{2\pi\hbar}\int V(x)e^{\mathrm{i}(p'-p)x/\hbar}\mathrm{d}x& \end{aligned} \tag{1.2}$$

1.2 力学量算符的矩阵表示

1.3 矩阵力学

1.4 幺正变换

1.5 Dirac 符号

1.6 线性谐振子及占有数表象

1.7 附录

1.7.1 动量表象能量本征方程推导

推导式(1.2). 由不进入表象的态矢量的能量本征方程出发

$$\hat{H}\left|\psi\right\rangle = (\hat{T}+V)\left|\psi\right\rangle = E\left|\psi\right\rangle \tag{1.3}$$

等式两边分别插入恒等算符

$$\int \left|p'\right\rangle\left\langle p'\right|\mathrm{d}p' = 1 \tag{1.4}$$

得到

$$\int (\hat{T}+V)\left|p'\right\rangle\left\langle p'|\psi\right\rangle \mathrm{d}p' = E\int \left|p'\right\rangle\left\langle p'|\psi\right\rangle \mathrm{d}p' \tag{1.5}$$

又 $\varphi(p') = \langle p'|\psi\rangle$ 得

$$\int (\hat{T}+V)\left|p'\right\rangle\varphi(p')\mathrm{d}p' = E\int \left|p'\right\rangle\varphi(p')\mathrm{d}p' \tag{1.6}$$

两边同时左乘 $\langle p|$ 得

$$\int \left\langle p\right|(\hat{T}+V)\left|p'\right\rangle\varphi(p')\mathrm{d}p' = E\int \left\langle p|p'\right\rangle\varphi(p')\mathrm{d}p' \tag{1.7}$$

$$\int \left\langle p\right|\hat{T}\left|p'\right\rangle\varphi(p')\mathrm{d}p' + \int \left\langle p\right|V\left|p'\right\rangle\varphi(p')\mathrm{d}p' = E\int \left\langle p|p'\right\rangle\varphi(p')\mathrm{d}p' \tag{1.8}$$

有克罗内克符号 $\delta_{pp'} = \langle p|p'\rangle$,$\hat{T}$ 的本征方程 $\hat{T}\left|p'\right\rangle = \frac{p^2}{2m}\left|p'\right\rangle$

$$\int \frac{p^2}{2m}\delta_{pp'}\varphi(p')\mathrm{d}p' + \int V_{pp'}\varphi(p')\mathrm{d}p' = E\int \delta_{pp'}\varphi(p')\mathrm{d}p' \tag{1.9}$$

由 δ 函数的挑选性 $\int \delta_{pp'}\varphi(p')\mathrm{d}p' = \varphi(p)$ 得

$$\boxed{\frac{p^2}{2m}\varphi(p) + \int V_{pp'}\varphi(p')\mathrm{d}p' = E\varphi(p)} \tag{1.10}$$

该式即为动量表象的能量本征方程.

其中 $V_{pp'}$

$$V_{pp'} = \left\langle p\right|V\left|p'\right\rangle \tag{1.11}$$

插入恒等算符

$$\int \left|x\right\rangle\left\langle x\right|\mathrm{d}x = 1 \tag{1.12}$$

得到

$$\begin{aligned}
V_{pp'} =& \langle p| V |p'\rangle \\
=& \iint \langle p|x\rangle \langle x| V |x'\rangle \langle x'|p'\rangle \mathrm{d}x\mathrm{d}x' \\
=& \iint \frac{1}{\sqrt{2\pi\hbar}} e^{-\mathrm{i}p'x/\hbar} V(x')\langle x|x'\rangle \frac{1}{\sqrt{2\pi\hbar}} e^{\mathrm{i}px/\hbar} \mathrm{d}x\mathrm{d}x' \\
=& \frac{1}{2\pi\hbar} \iint e^{-\mathrm{i}p'x/\hbar} V(x')\delta_{xx'} e^{\mathrm{i}px/\hbar} \mathrm{d}x\mathrm{d}x' \\
=& \frac{1}{2\pi\hbar} \int e^{-\mathrm{i}p'x/\hbar} V(x) e^{\mathrm{i}px/\hbar} \mathrm{d}x \\
=& \frac{1}{2\pi\hbar} \int V(x) e^{-\mathrm{i}(p'-p)x/\hbar} \mathrm{d}x
\end{aligned} \tag{1.13}$$

所以

$$\boxed{V_{pp'} = \frac{1}{2\pi\hbar} \int V(x) e^{-\mathrm{i}(p'-p)x/\hbar} \mathrm{d}x} \tag{1.14}$$